

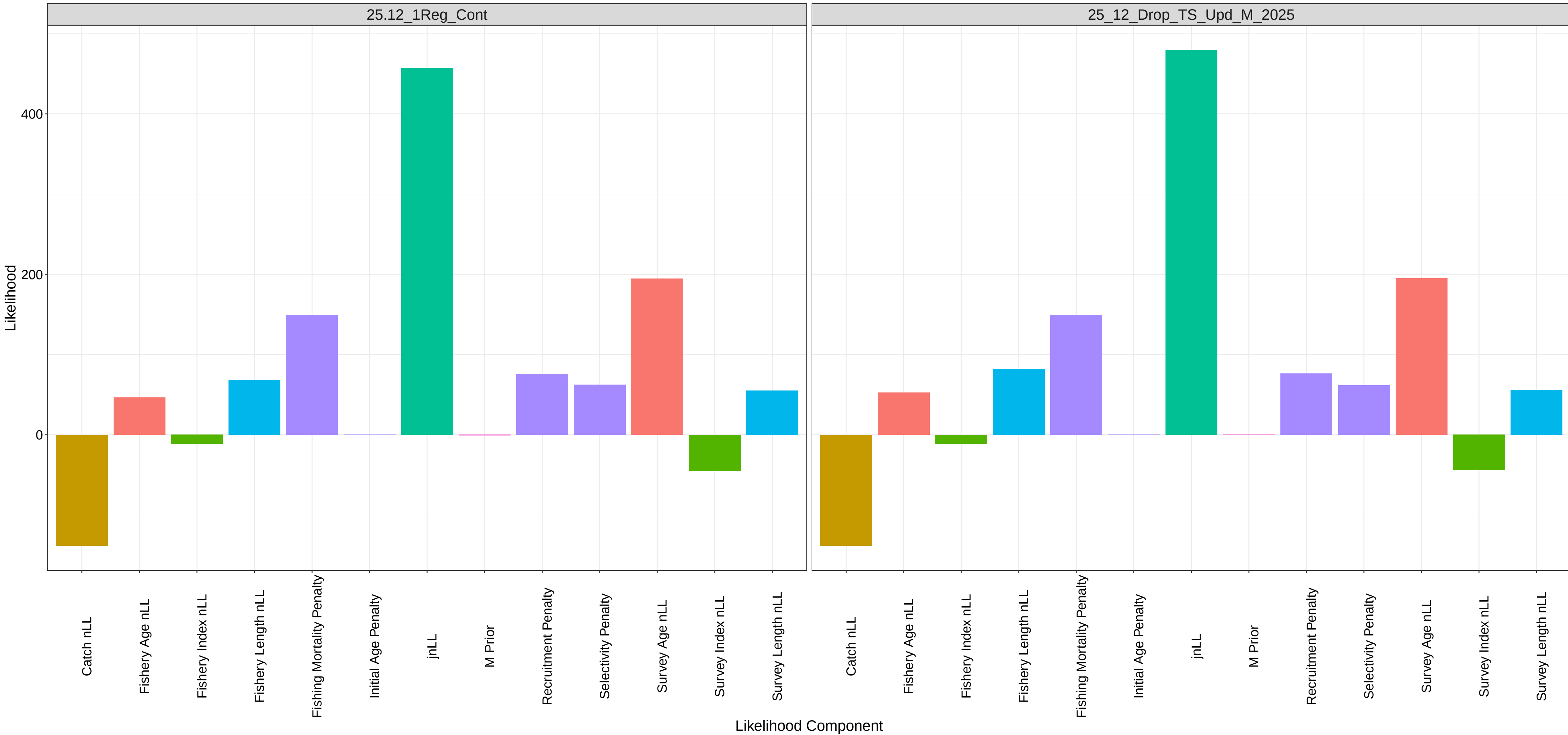
|   | Model                    | Region | Terminal_SSB | Terminal_SSB0 | Terminal_F | Catch_Advice | B_Ref_Pt | F_Ref_Pt | B_over_B_Ref | B_over_DynB_Ref | F_over_F_Ref |
|---|--------------------------|--------|--------------|---------------|------------|--------------|----------|----------|--------------|-----------------|--------------|
| 1 | 25.12_1Reg_Cont          | 1      | 206.6999     | 357.0208      | 0.04784668 | 39.55138     | 131.4549 | 0.07875  | 1.57240      | 0.57896         | 0.60758      |
| 2 | 25_12_Drop_TS_Upd_M_2025 | 1      | 220.3244     | 357.0208      | 0.04360258 | 43.80200     | 130.3881 | 0.08332  | 1.68976      | 0.61712         | 0.52334      |

|   | Model                    | jnLL     | M Prior    | Recruitment Penalty | Selectivity Penalty | Catch nLL | Fishing Mortality Penalty | Survey Index nLL | Fishery Index nLL | Initial Age Penalty | Survey Age nLL | Fishery Age nLL | Survey Length nLL | Fishery Length nLL |
|---|--------------------------|----------|------------|---------------------|---------------------|-----------|---------------------------|------------------|-------------------|---------------------|----------------|-----------------|-------------------|--------------------|
| 1 | 25.12_1Reg_Cont          | 456.8247 | -1.0189779 | 75.86092            | 62.56423            | -138.3008 | 149.3833                  | -45.39642        | -11.21223         | 0.2741517           | 194.6365       | 46.66608        | 55.19863          | 68.16928           |
| 2 | 25_12_Drop_TS_Upd_M_2025 | 479.6137 | -0.1800847 | 76.16572            | 61.75755            | -138.3725 | 149.3287                  | -44.29805        | -10.93230         | 0.3075879           | 195.1024       | 52.58410        | 55.96482          | 82.18583           |

# Negative Log-Likelihood

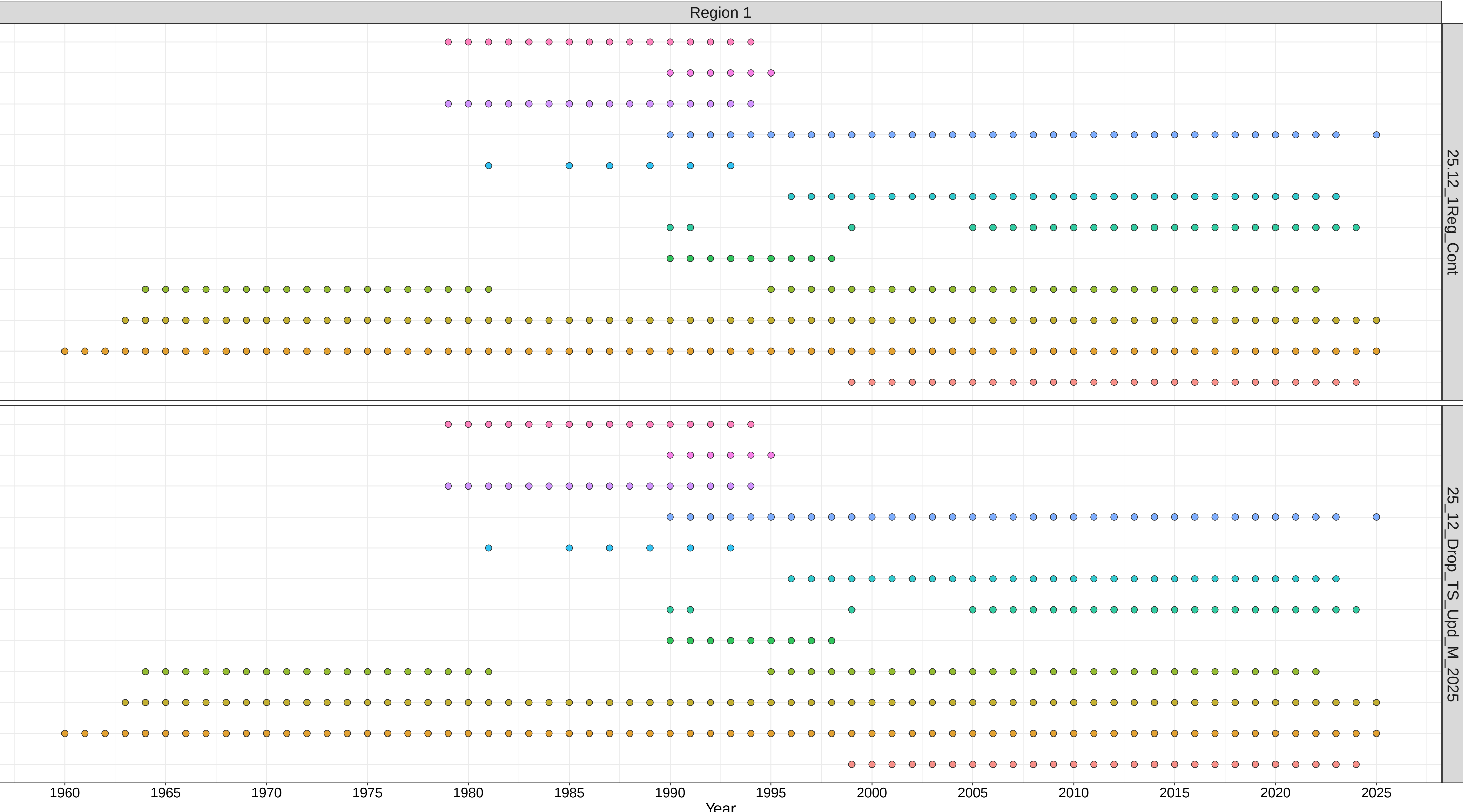
Type

- Age
- Catch
- Index
- jnLL
- Length
- Penalty
- Prior



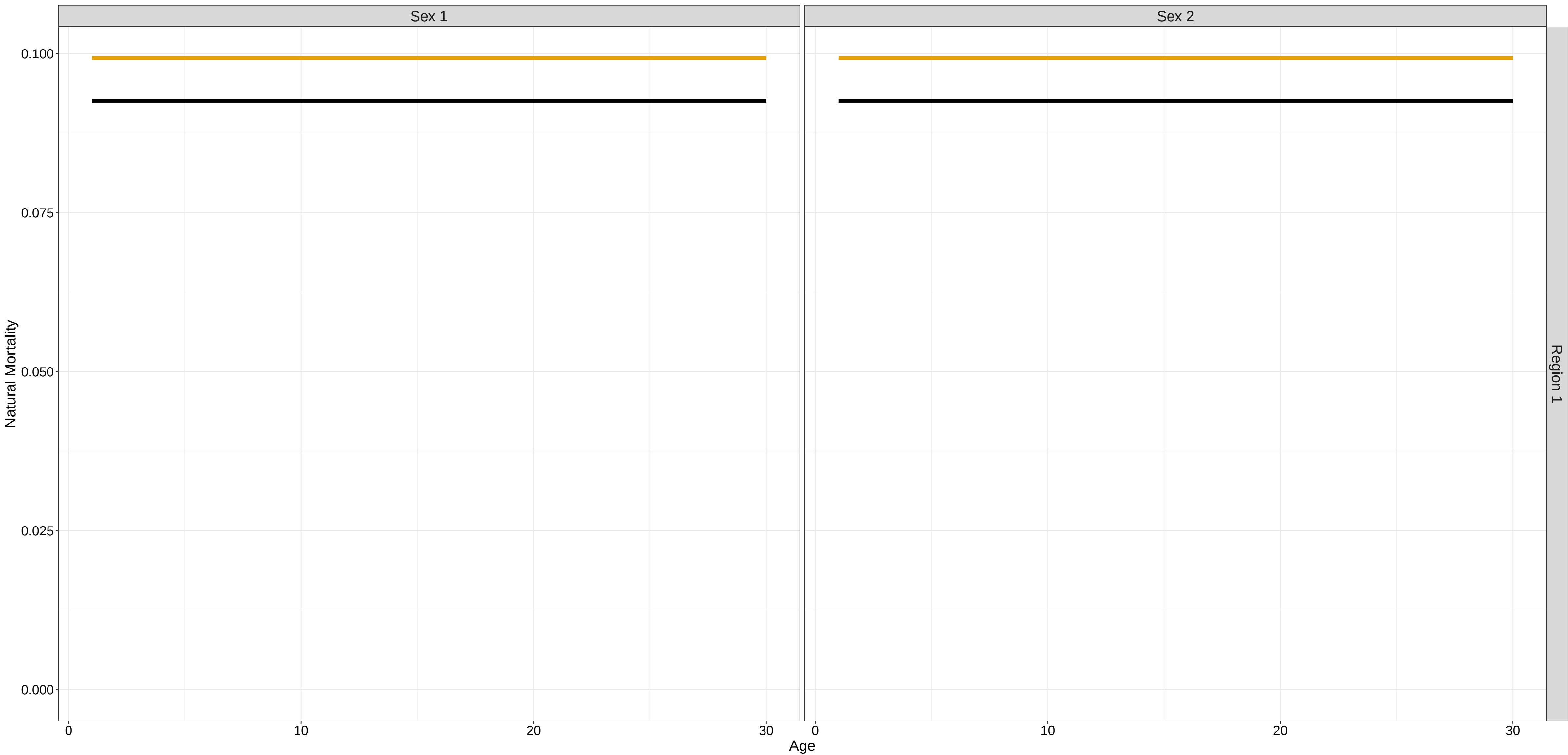


Data



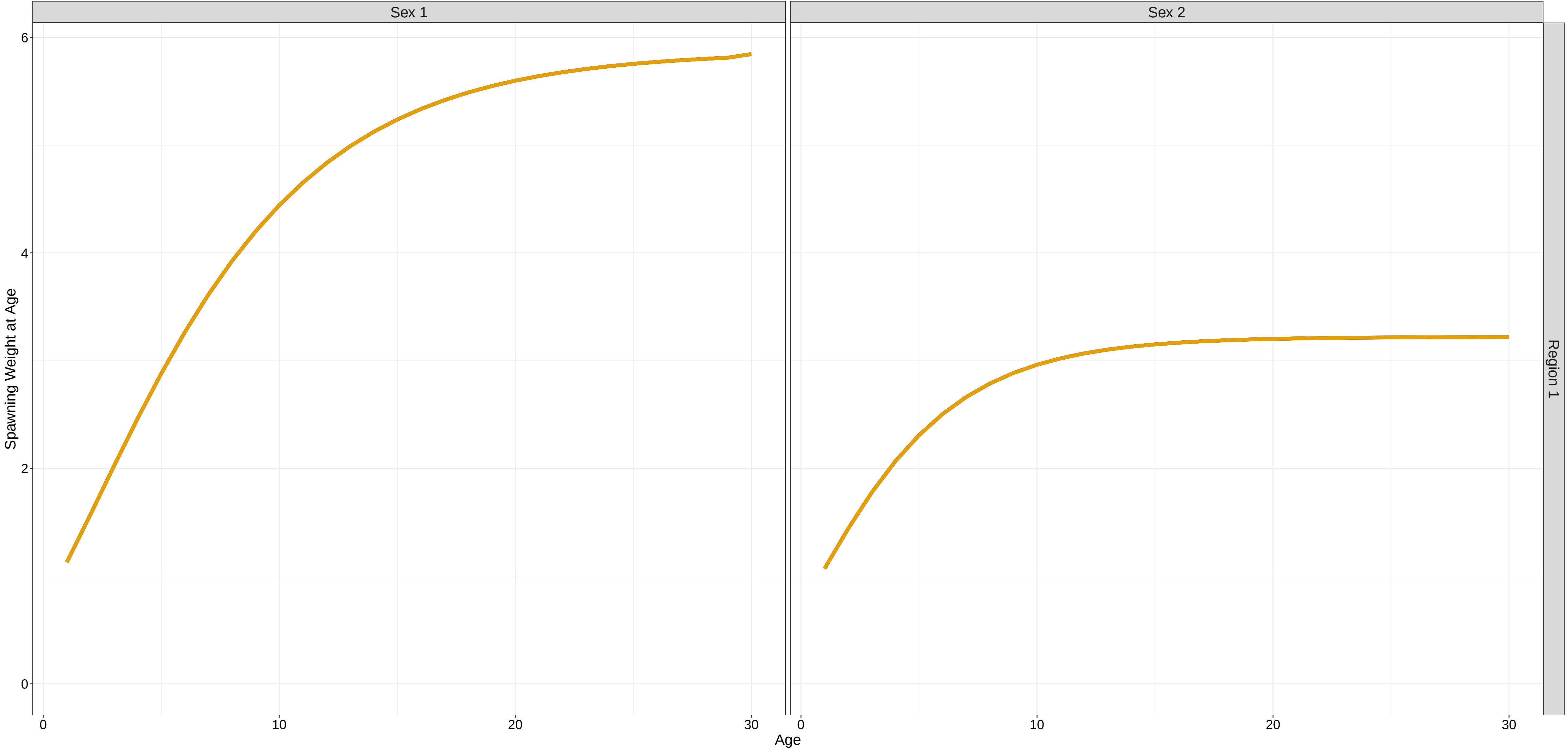
# Natural Mortality

Model 25.12\_1Reg\_Cont 25\_12\_Drop\_TS\_Upd\_M\_2025



Weight-at-Age

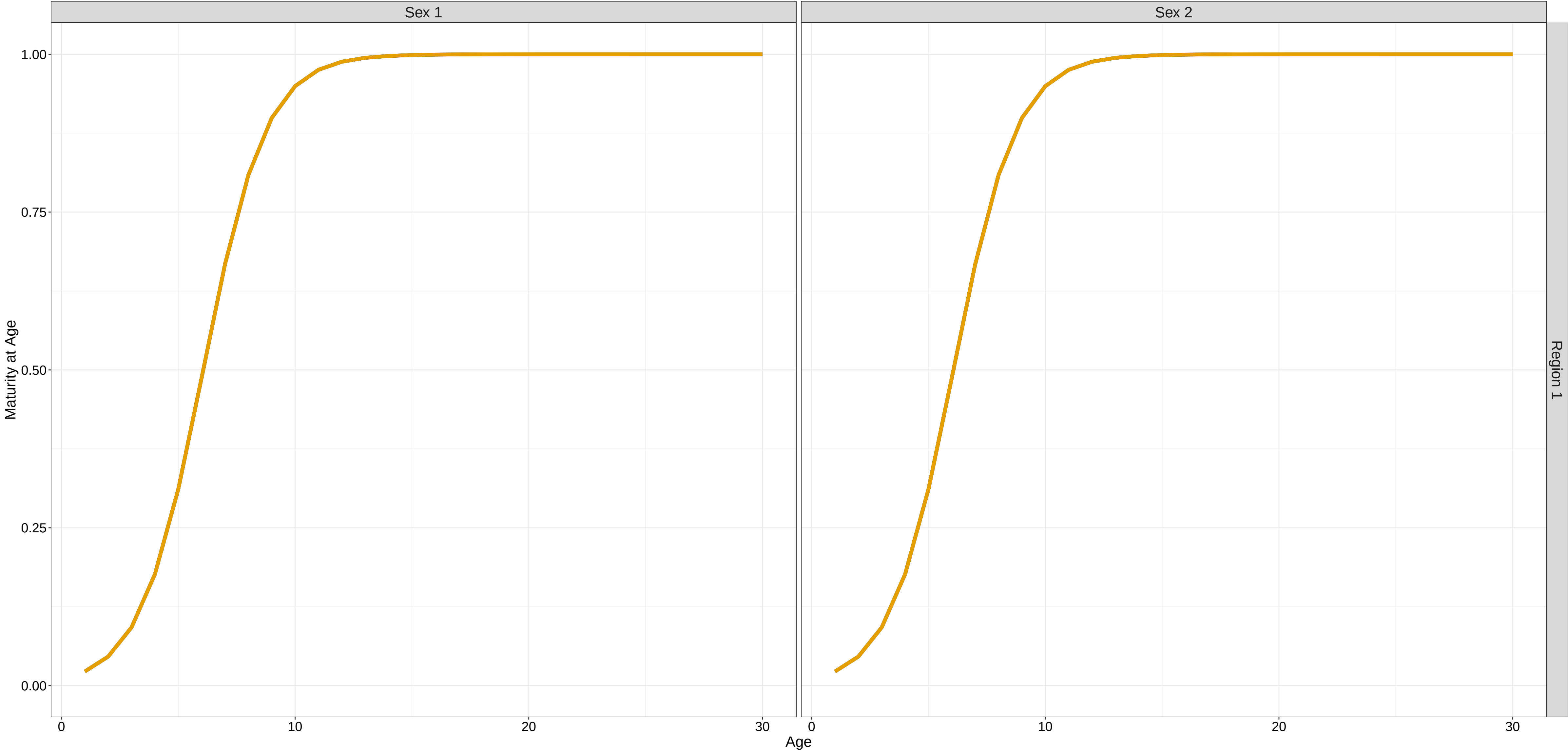
Model 25.12\_1Reg\_Cont 25\_12\_Drop\_TS\_Upd\_M\_2025





# Maturity

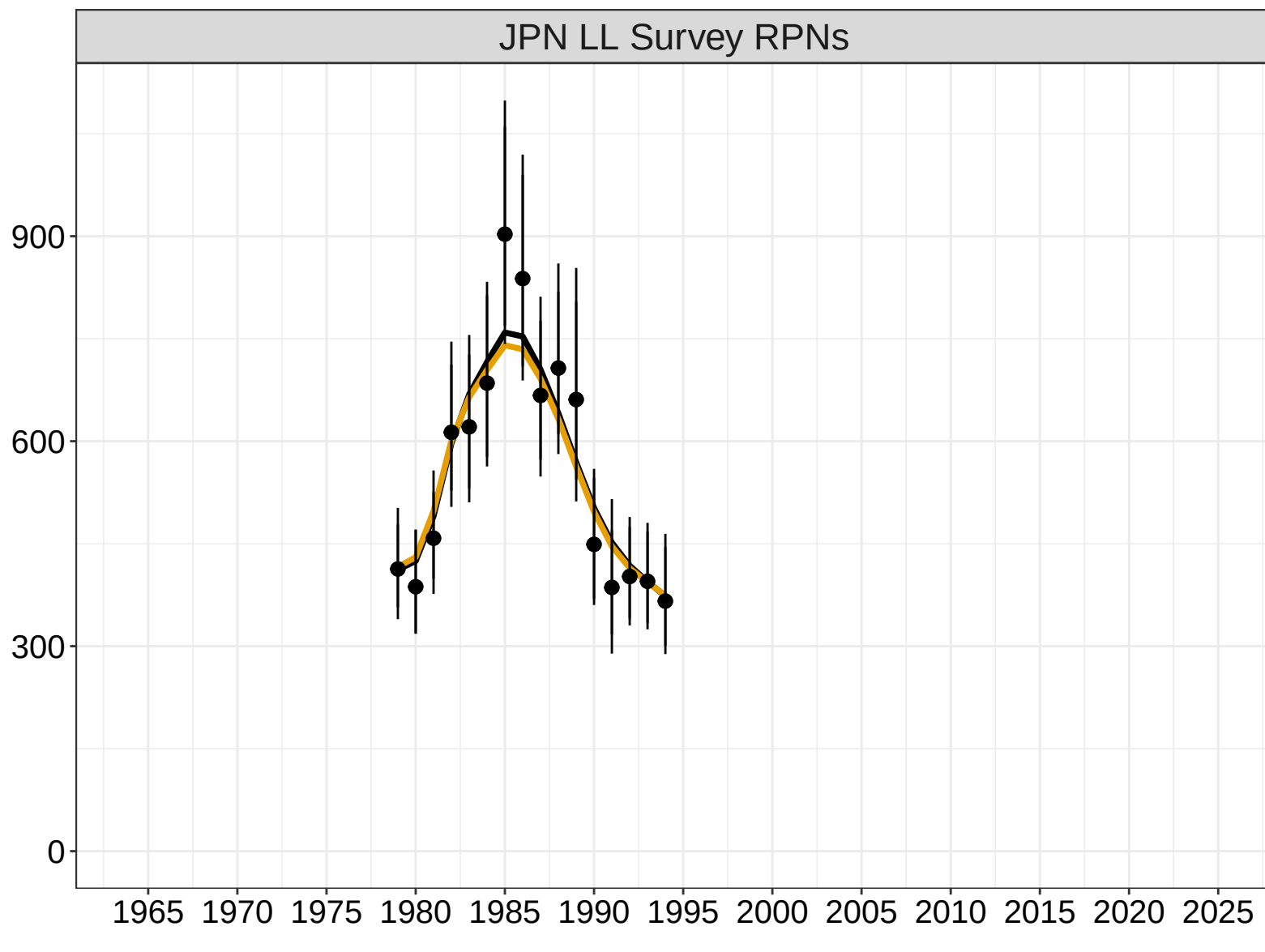
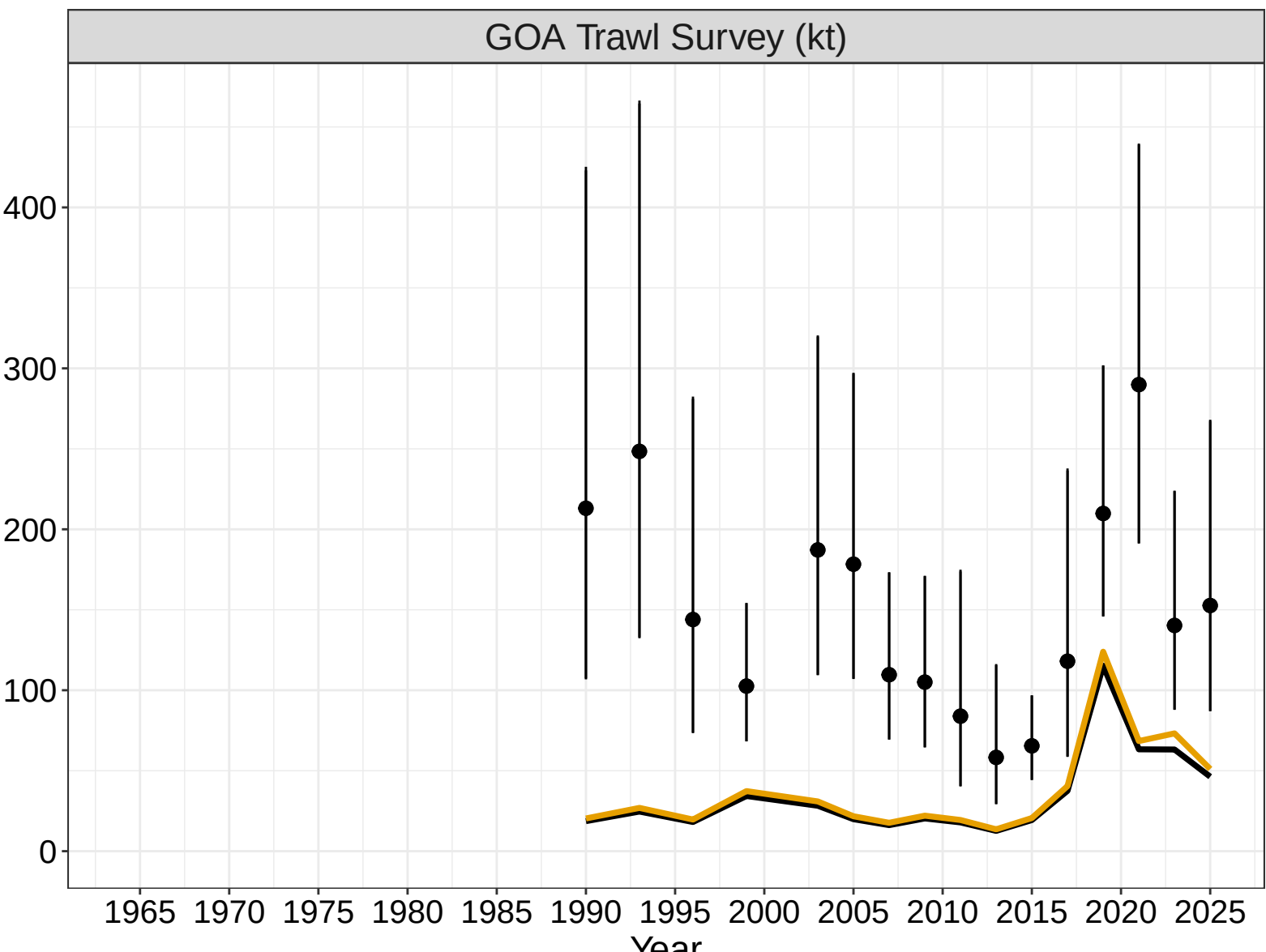
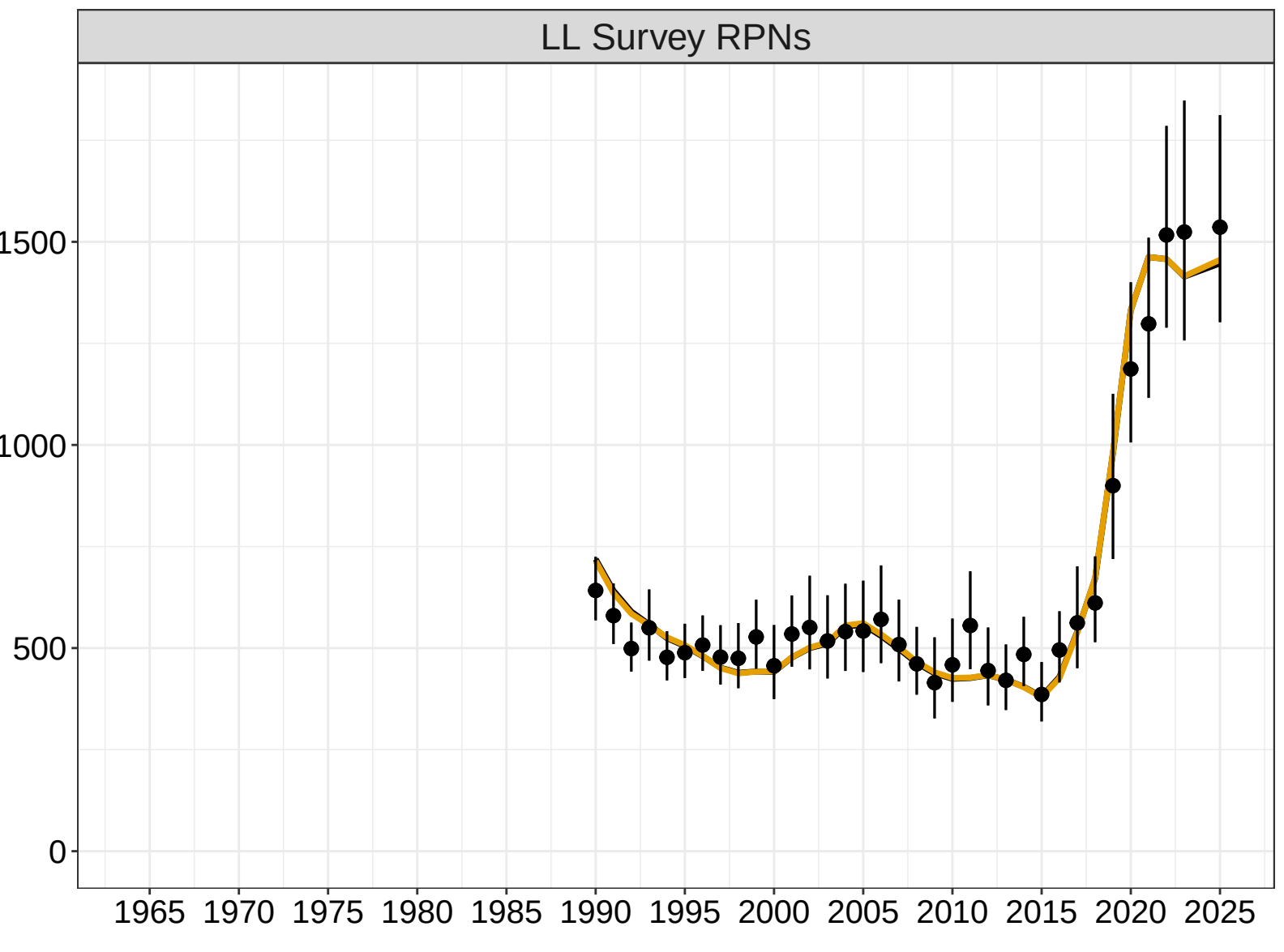
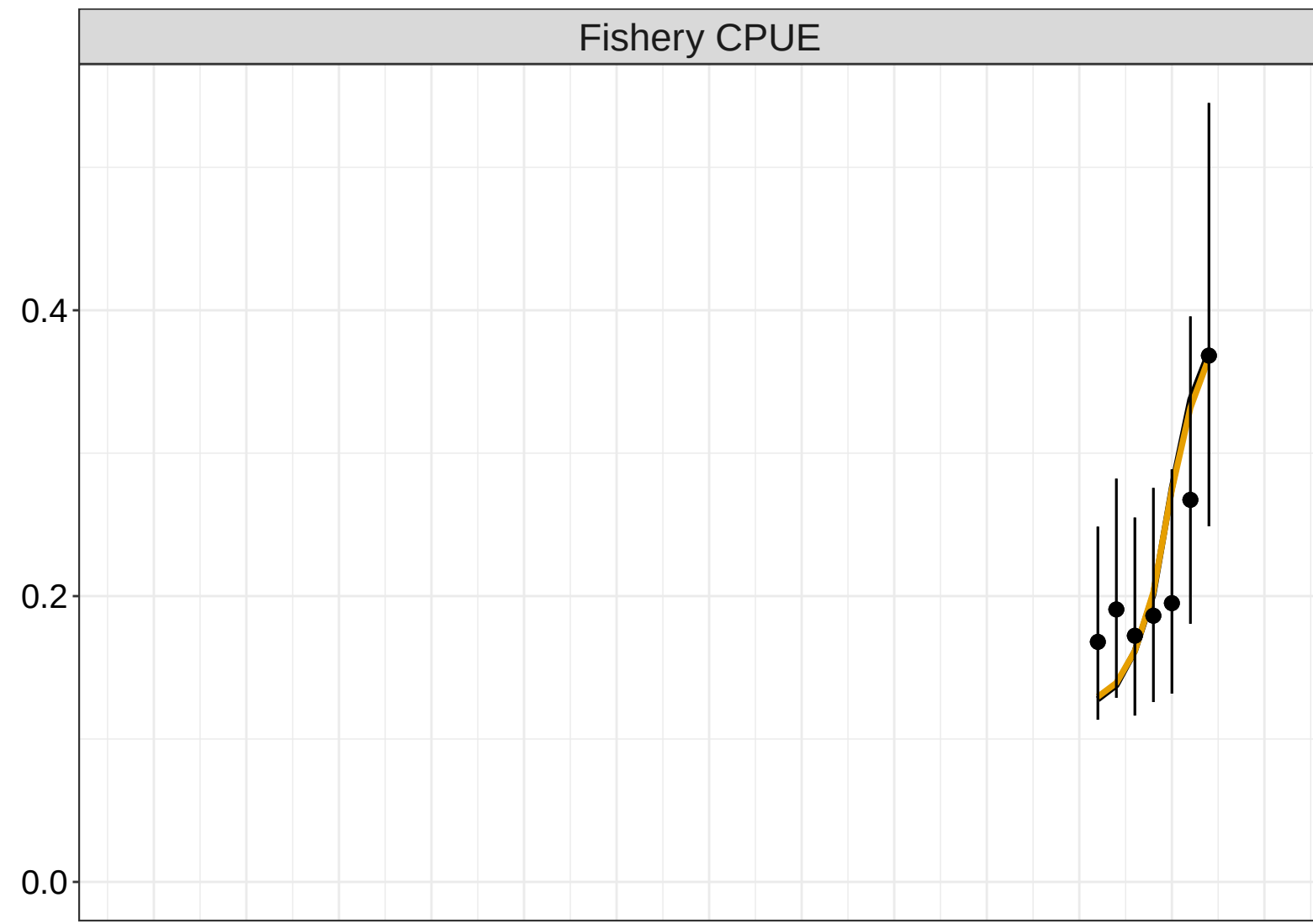
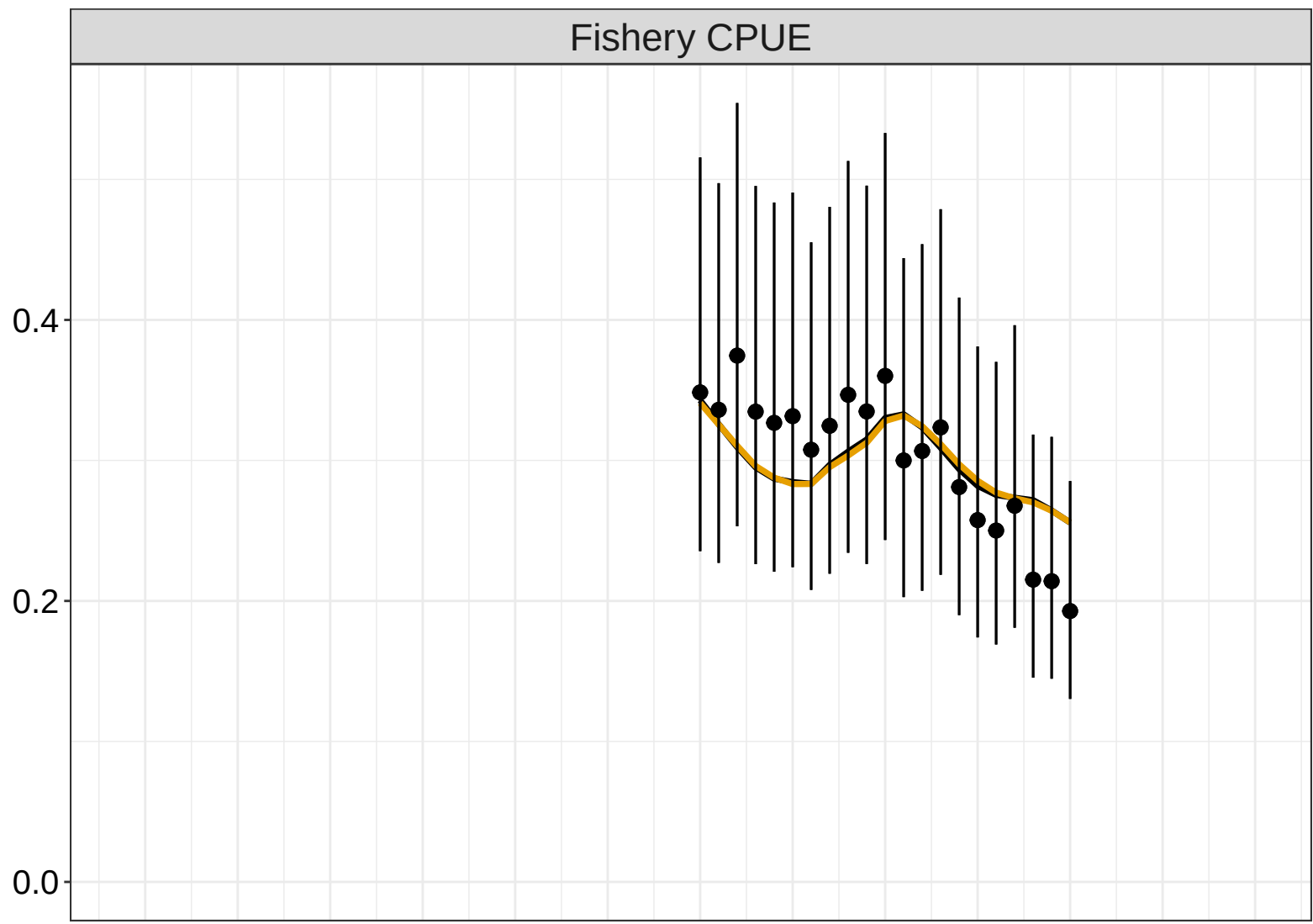
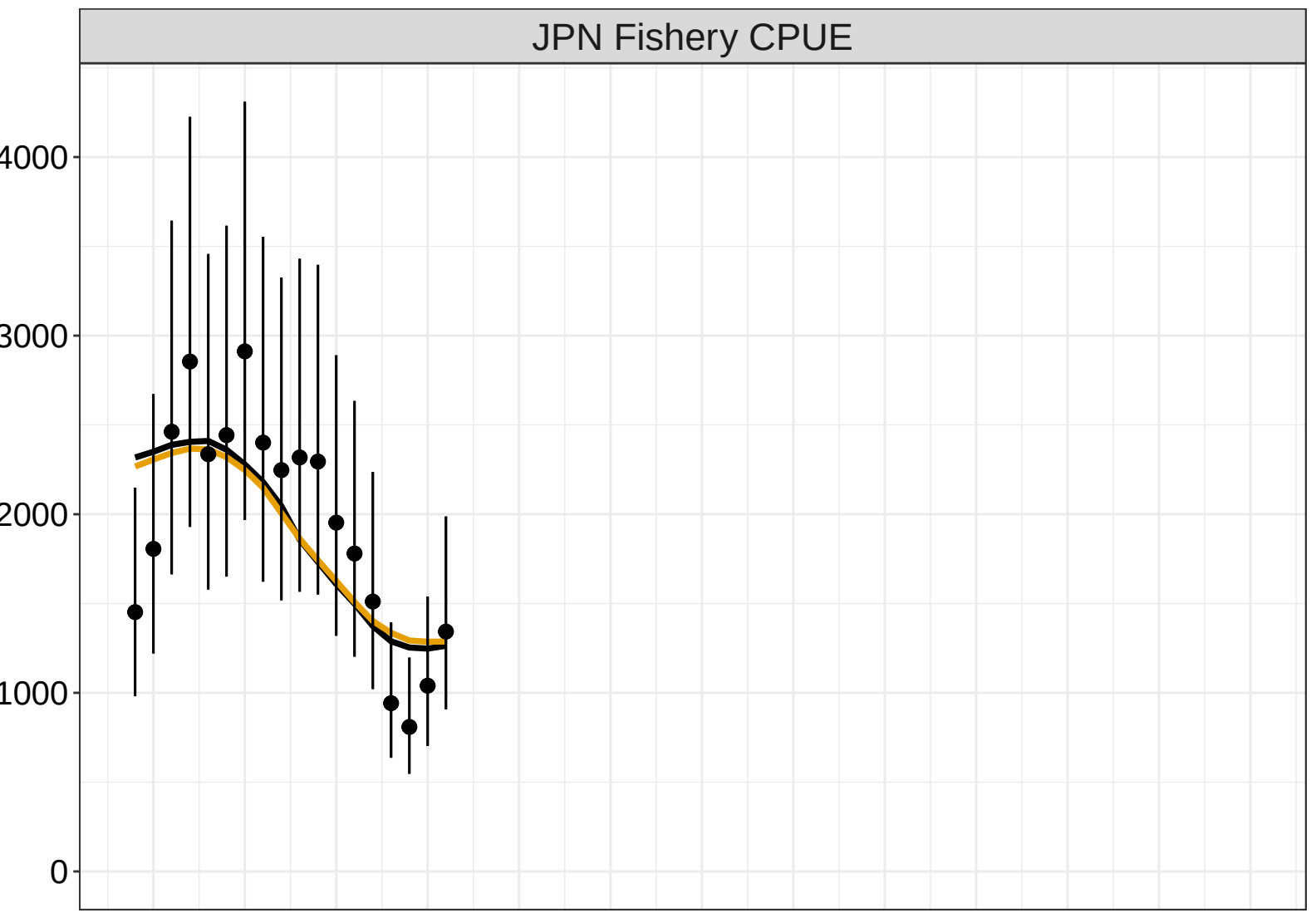
Model 25.12\_1Reg\_Cont 25\_12\_Drop\_TS\_Upd\_M\_2025



# Index Fits

Model — 25.12\_1Reg\_Cont — 25\_12\_Drop\_TS\_Upd\_M\_2025

Index

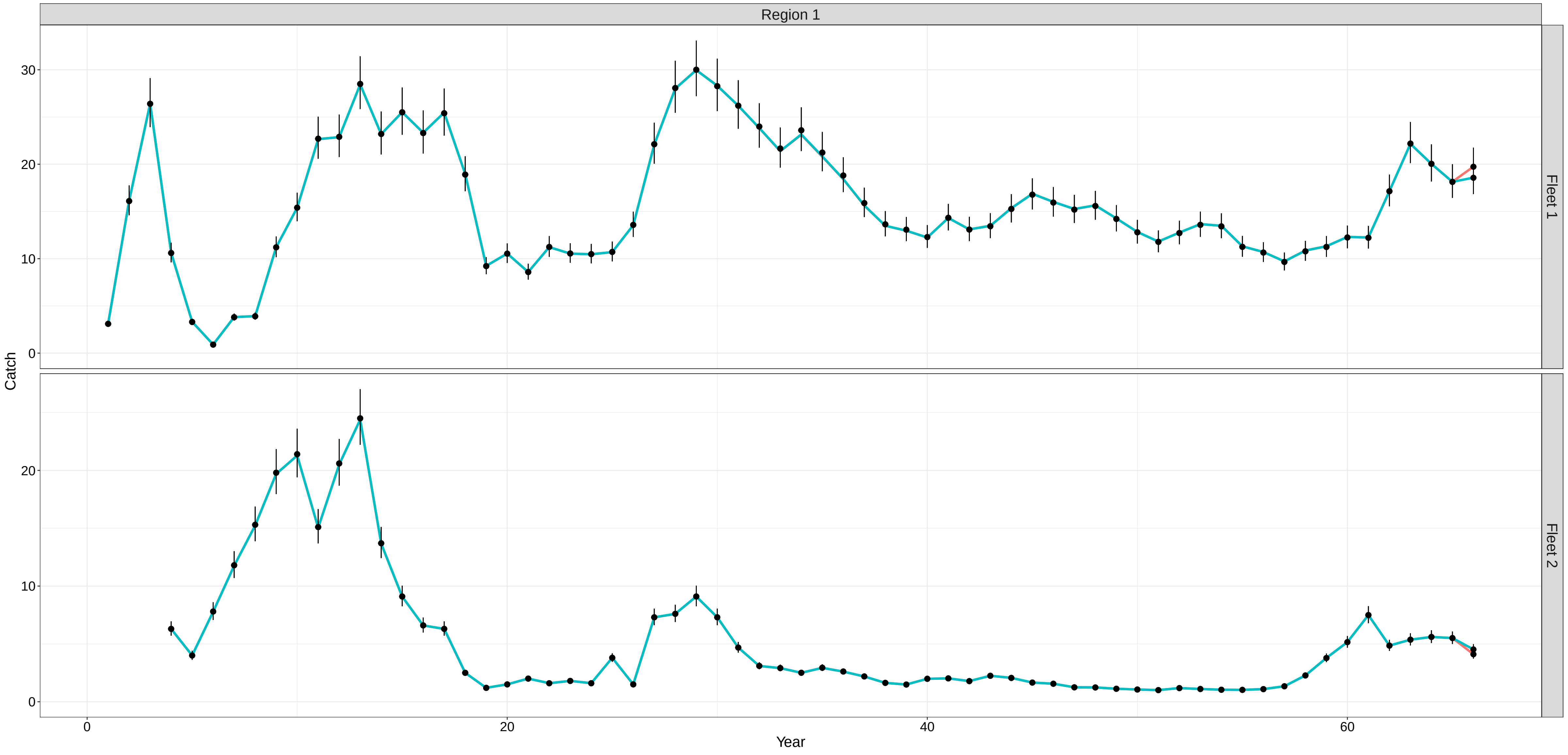


Year



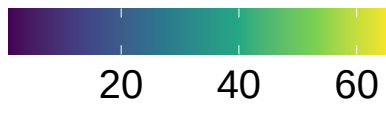
Catch

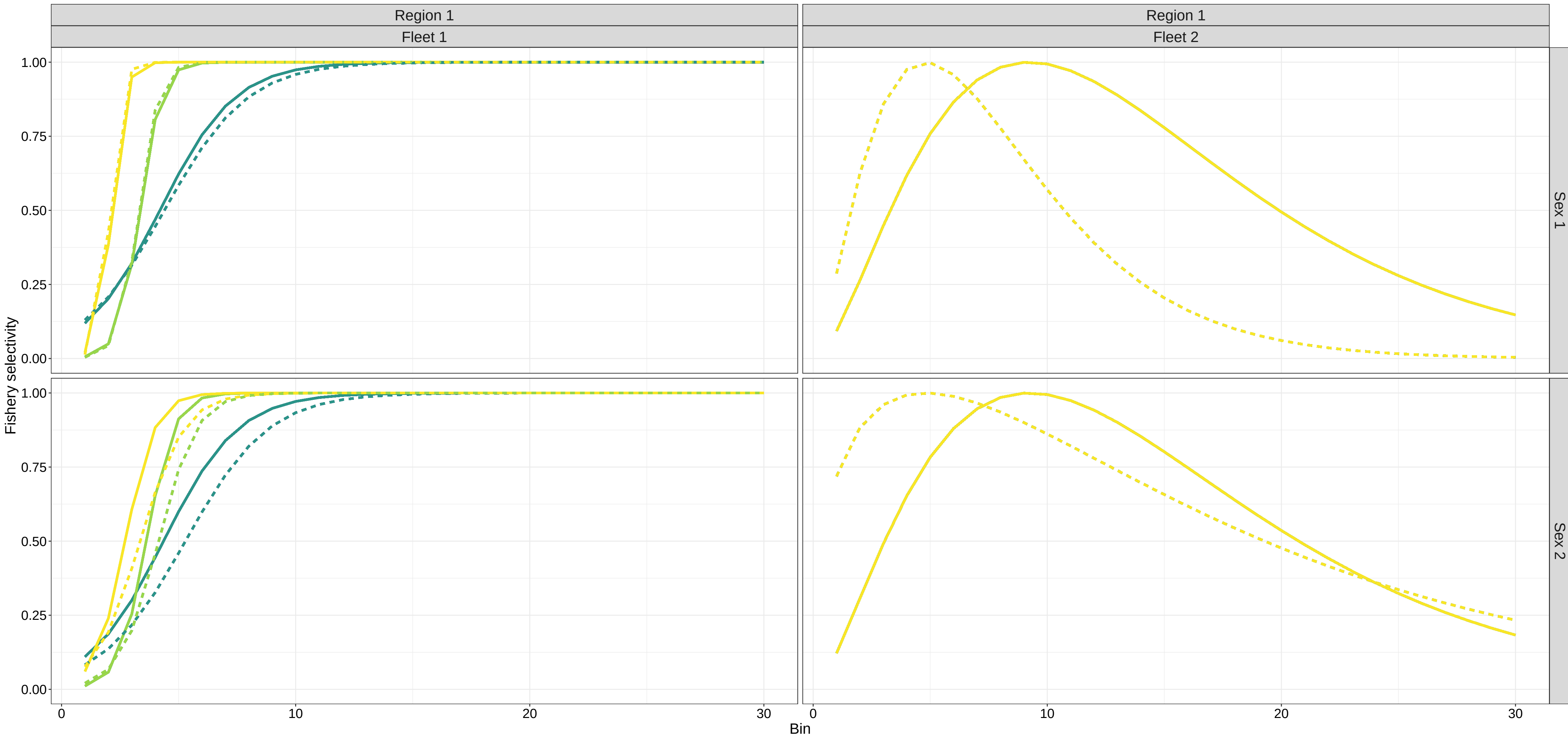
Model 25.12\_1Reg\_Cont 25\_12\_Drop\_TS\_Upd\_M\_2025



# Fishery Selectivity

Model — 25.12\_1Reg\_Cont - - - 25\_12\_Drop\_TS\_Upd\_M\_2025

Year  20 40 60



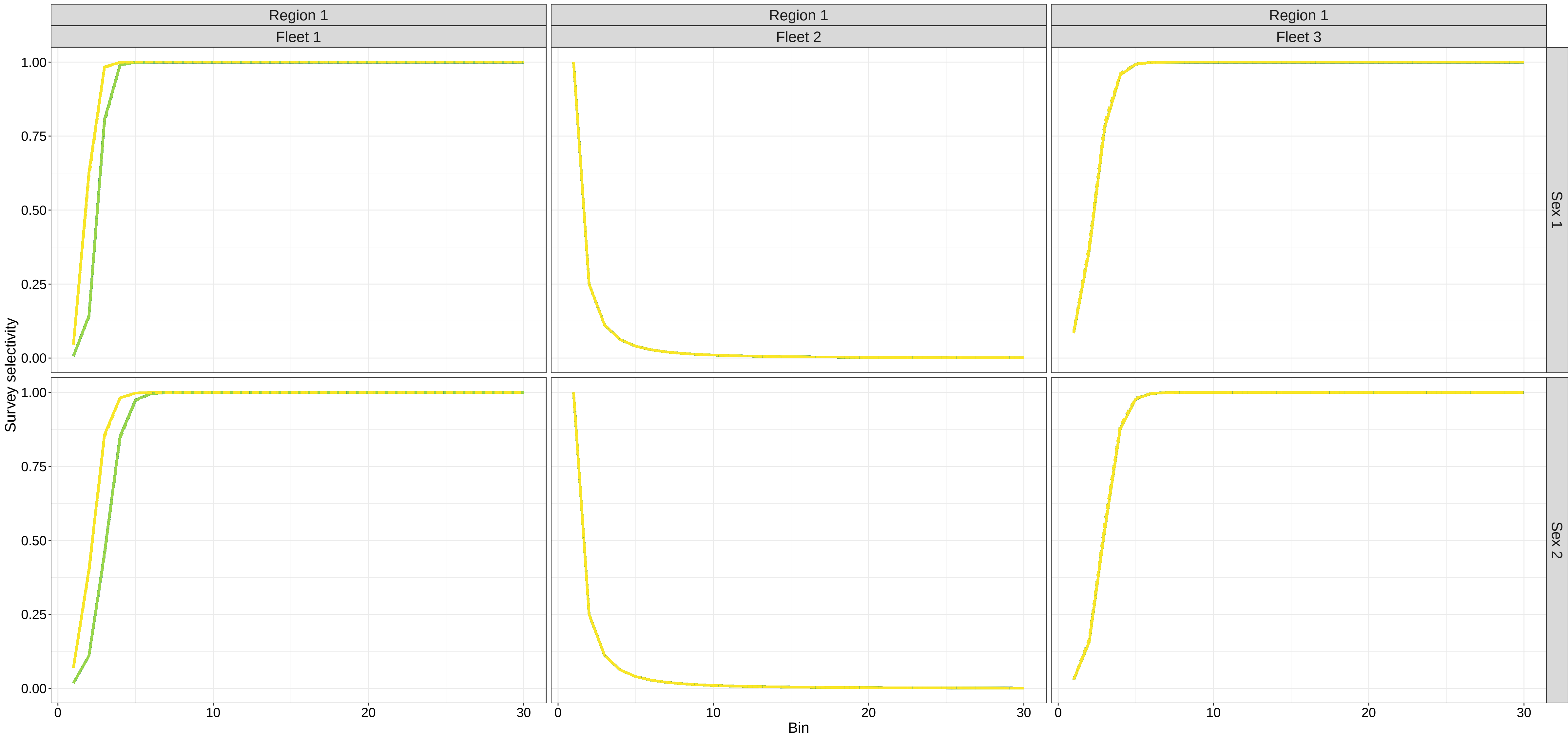
# Survey Selectivity

Model

— 25.12\_1Reg\_Cont

- - - 25\_12\_Drop\_TS\_Upd\_M\_2025

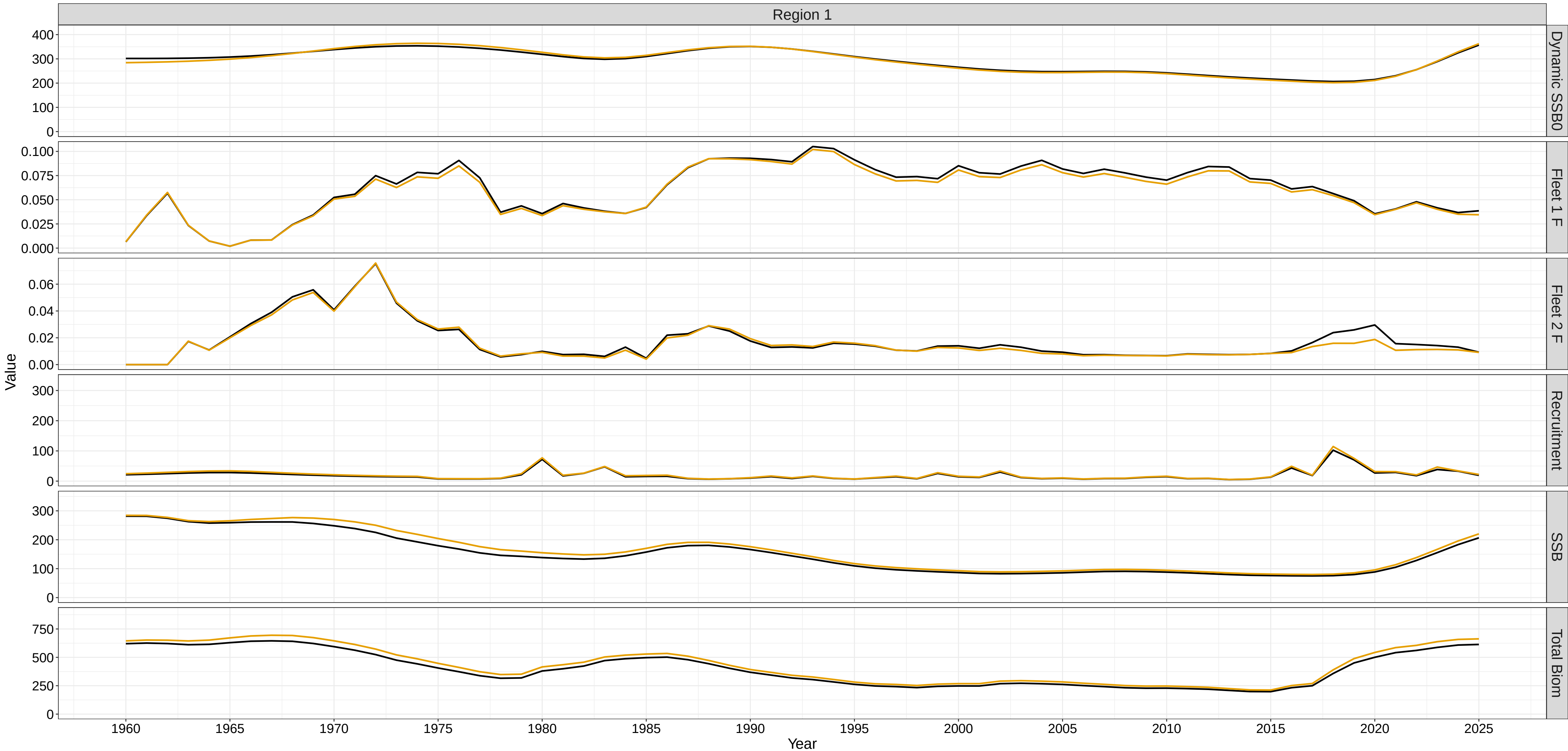
Year





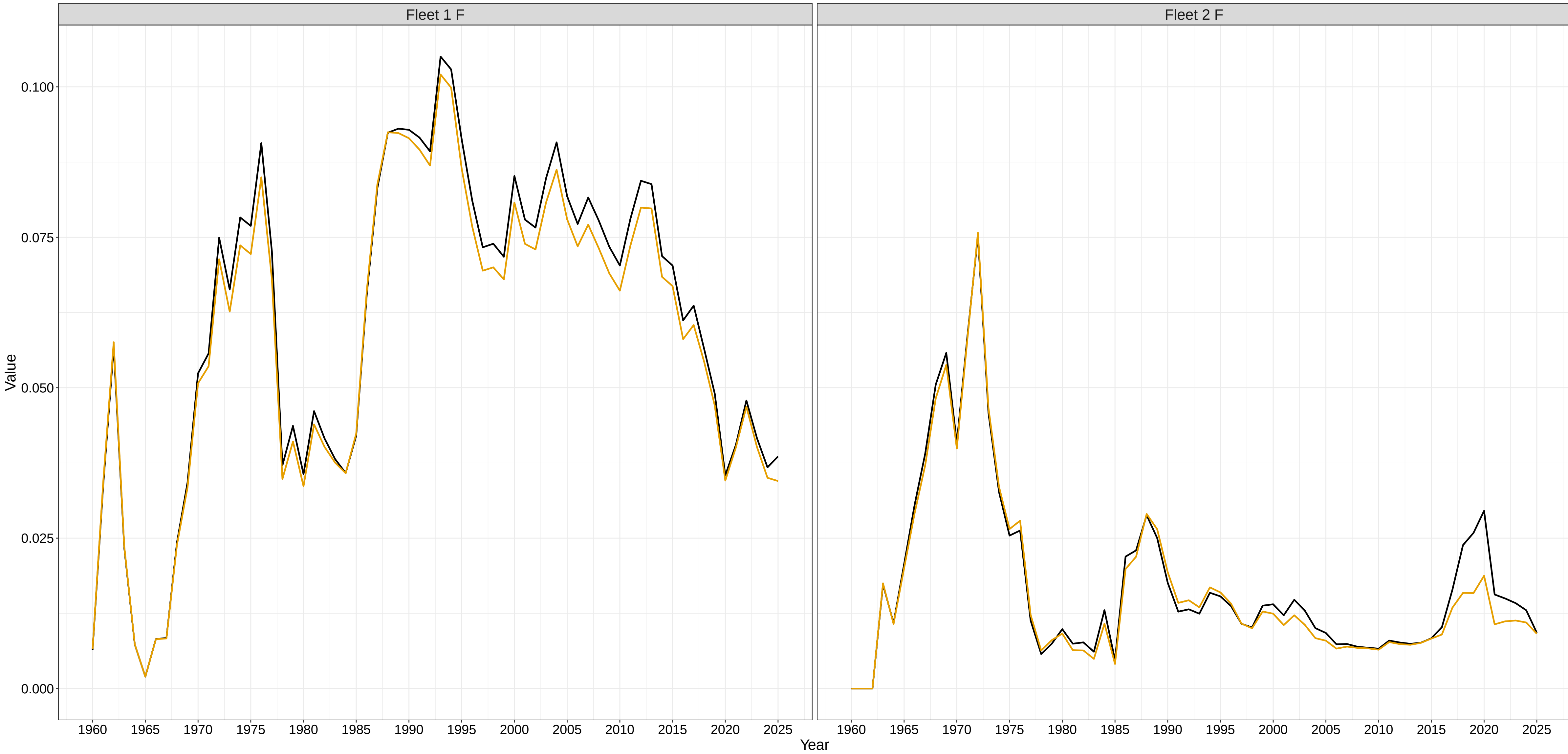
# Time Series Plots

Model — 25.12\_1Reg\_Cont — 25\_12\_Drop\_TS\_Upd\_M\_2025



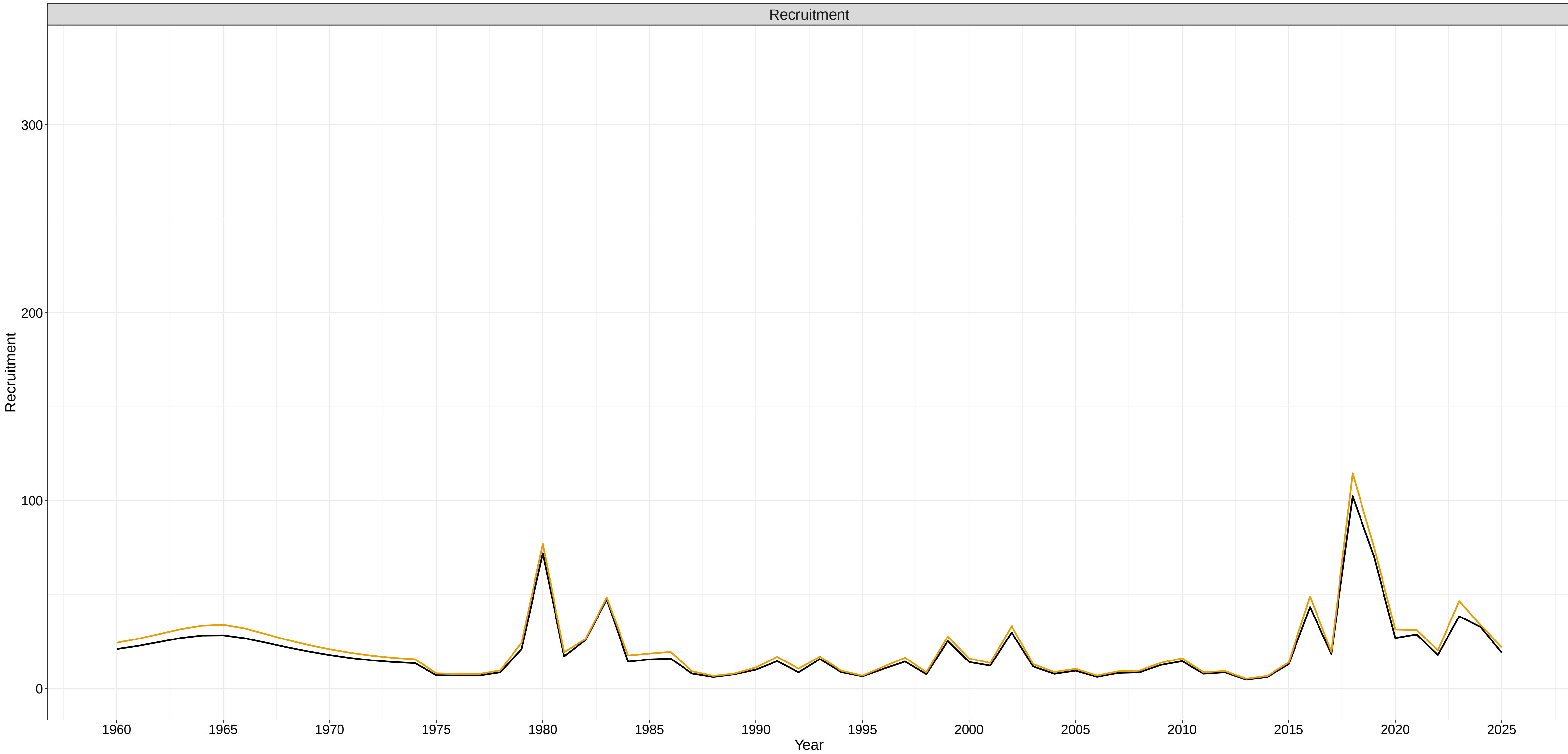
# Fishing Mortality

Model — 25.12\_1Reg\_Cont — 25\_12\_Drop\_TS\_Upd\_M\_2025



# Recruitment

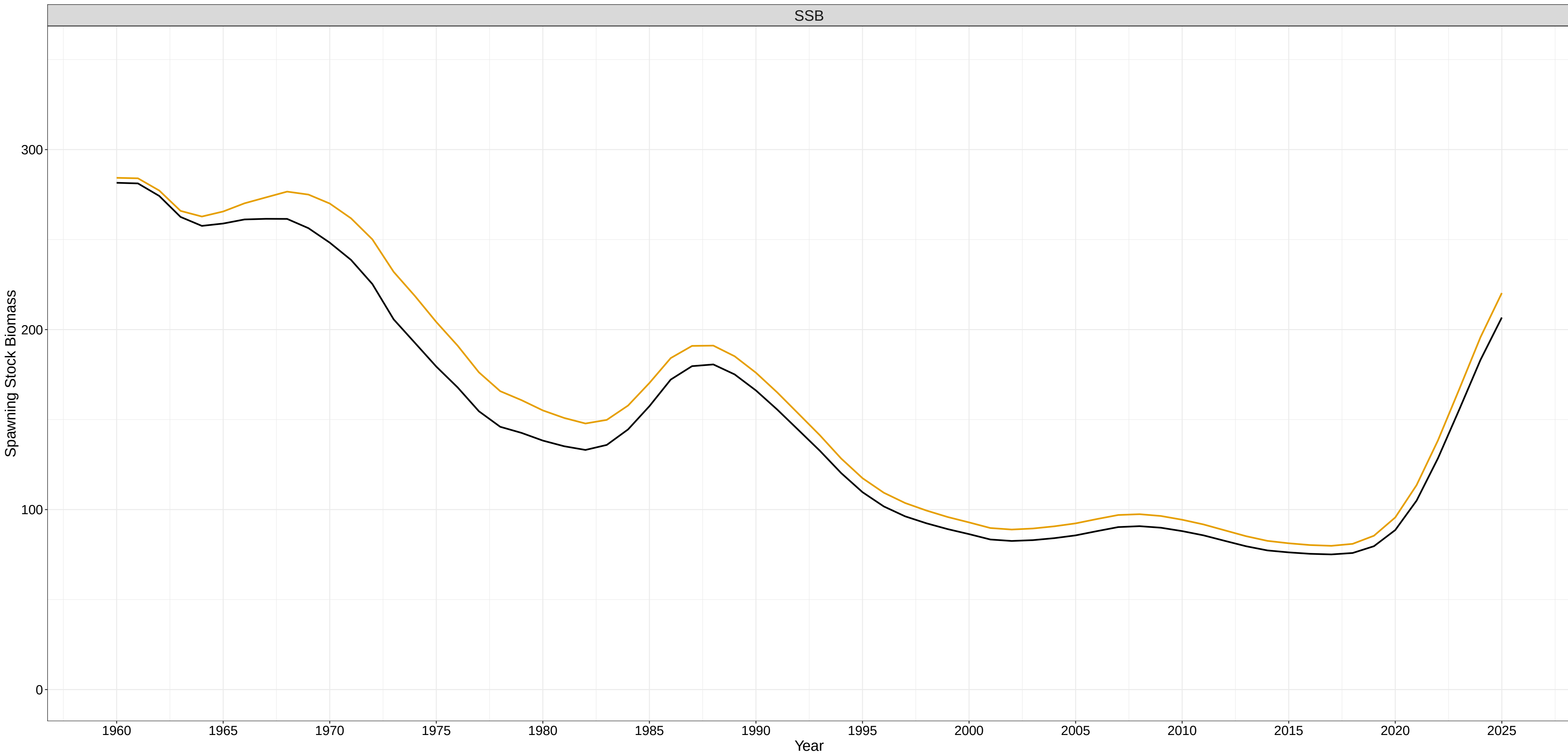
Model — 25.12\_1Reg\_Cont — 25\_12\_Drop\_TS\_Upd\_M\_2025





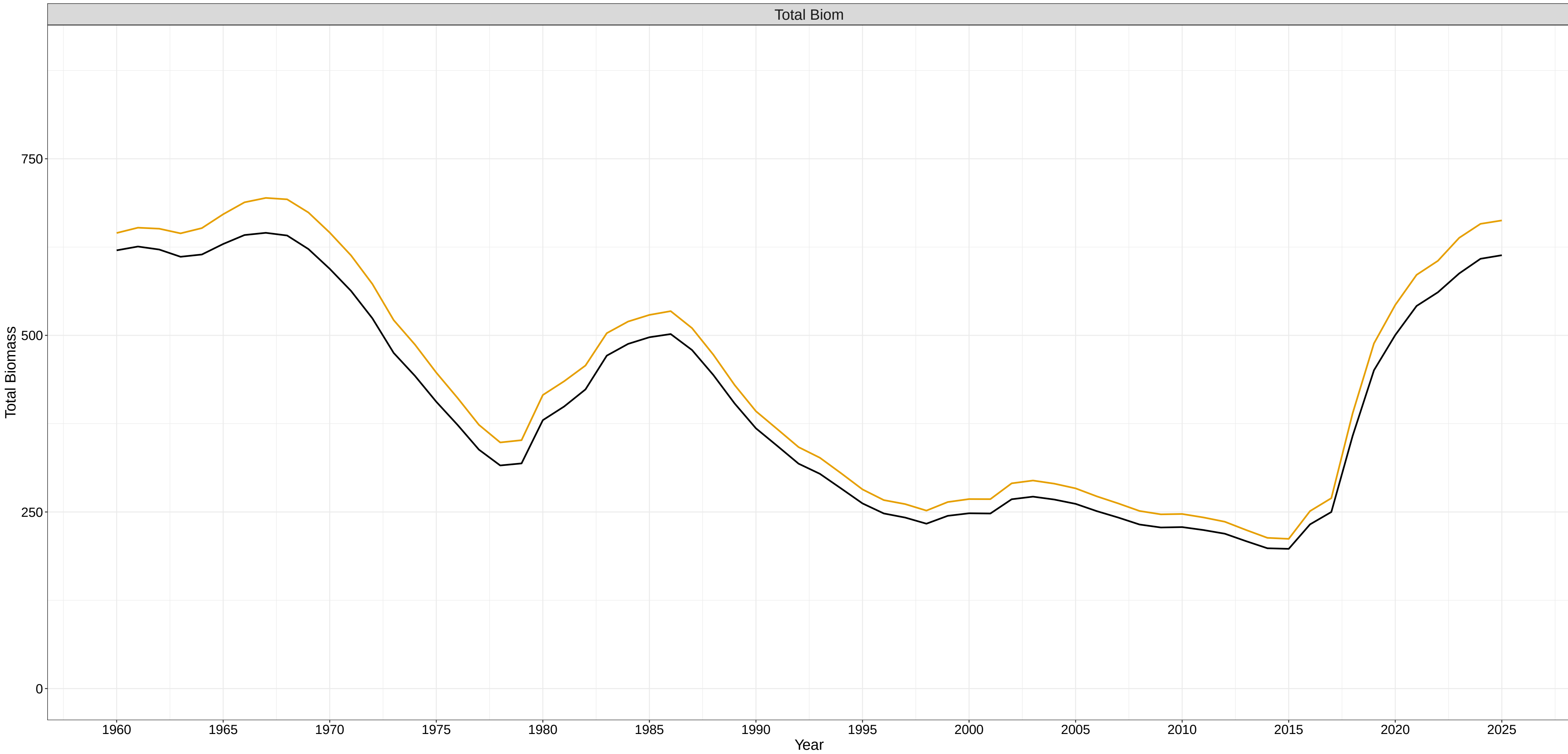
SSB

Model — 25.12\_1Reg\_Cont — 25\_12\_Drop\_TS\_Upd\_M\_2025



# Biomass

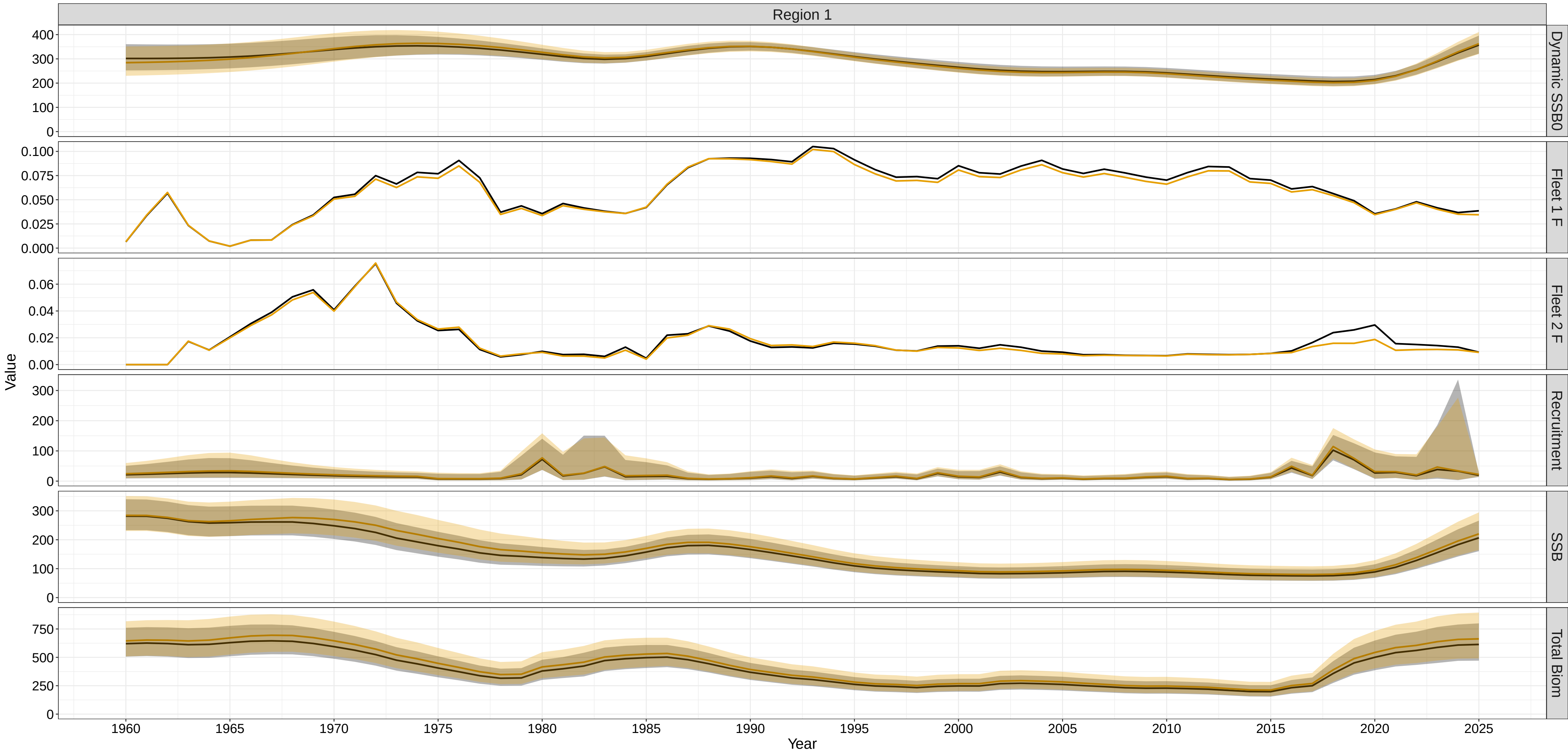
Model — 25.12\_1Reg\_Cont — 25\_12\_Drop\_TS\_Upd\_M\_2025





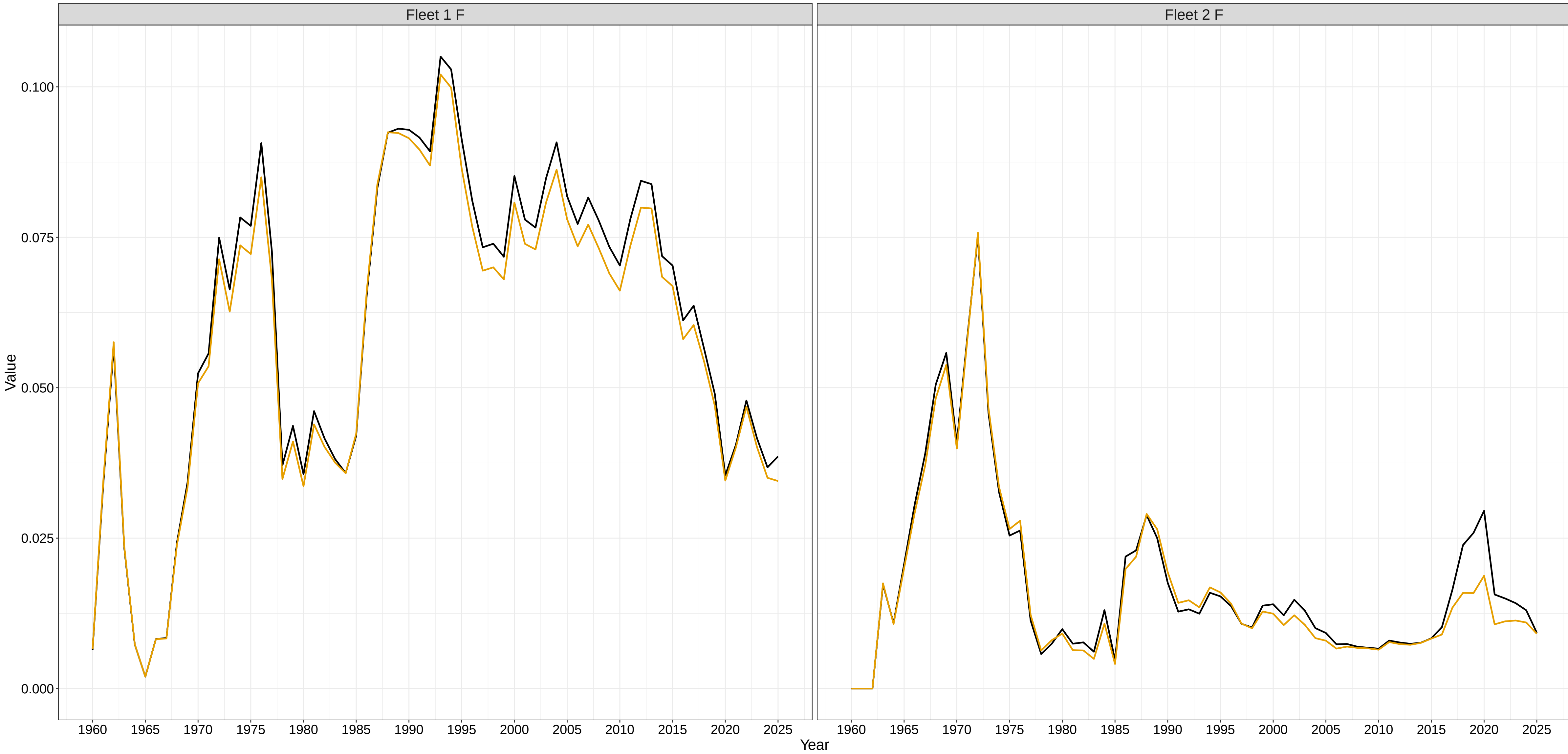
# Time Series Plots

Model 25.12\_1Reg\_Cont 25\_12\_Drop\_TS\_Upd\_M\_2025

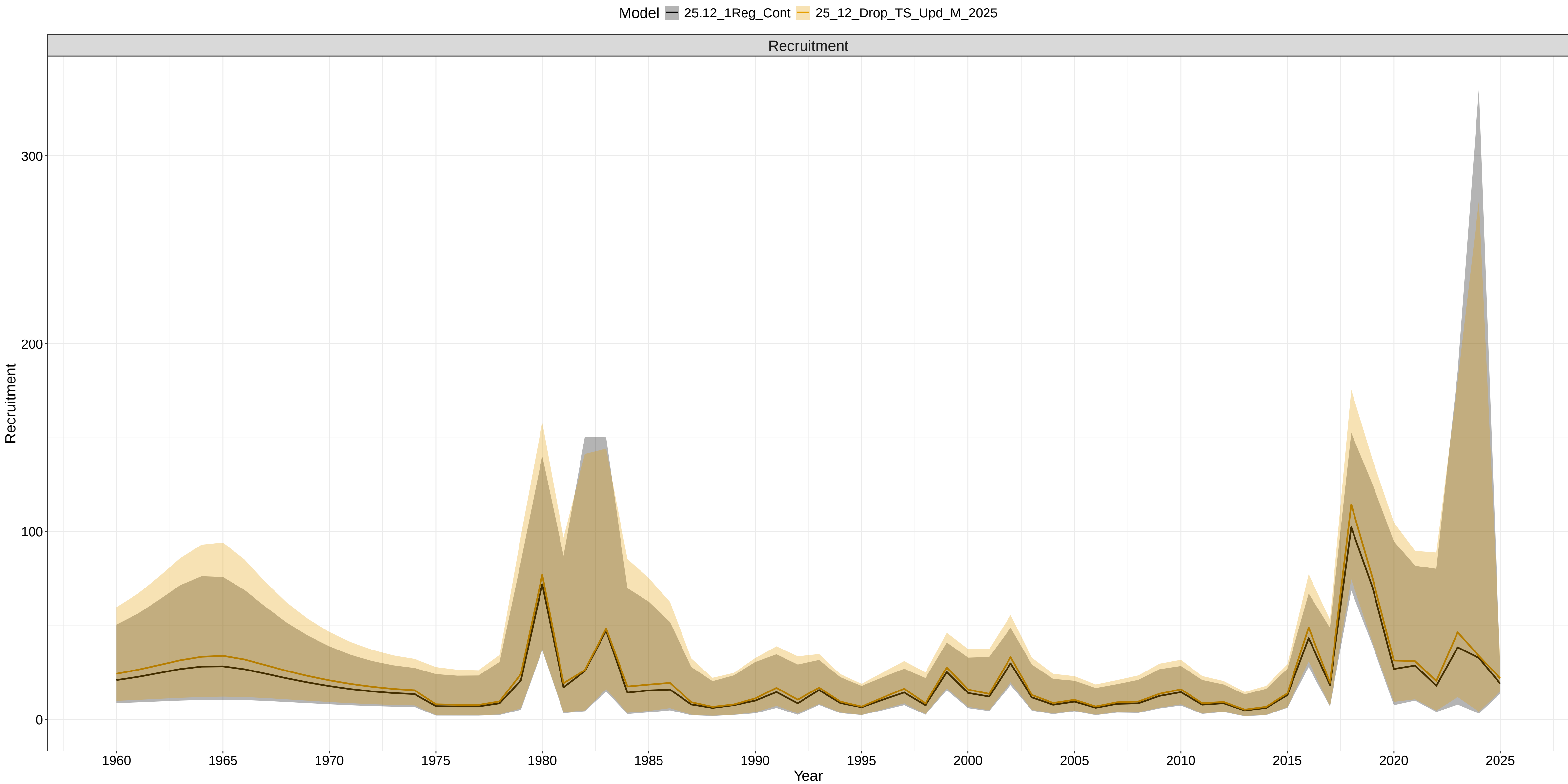


# Fishing Mortality

Model — 25.12\_1Reg\_Cont — 25\_12\_Drop\_TS\_Upd\_M\_2025



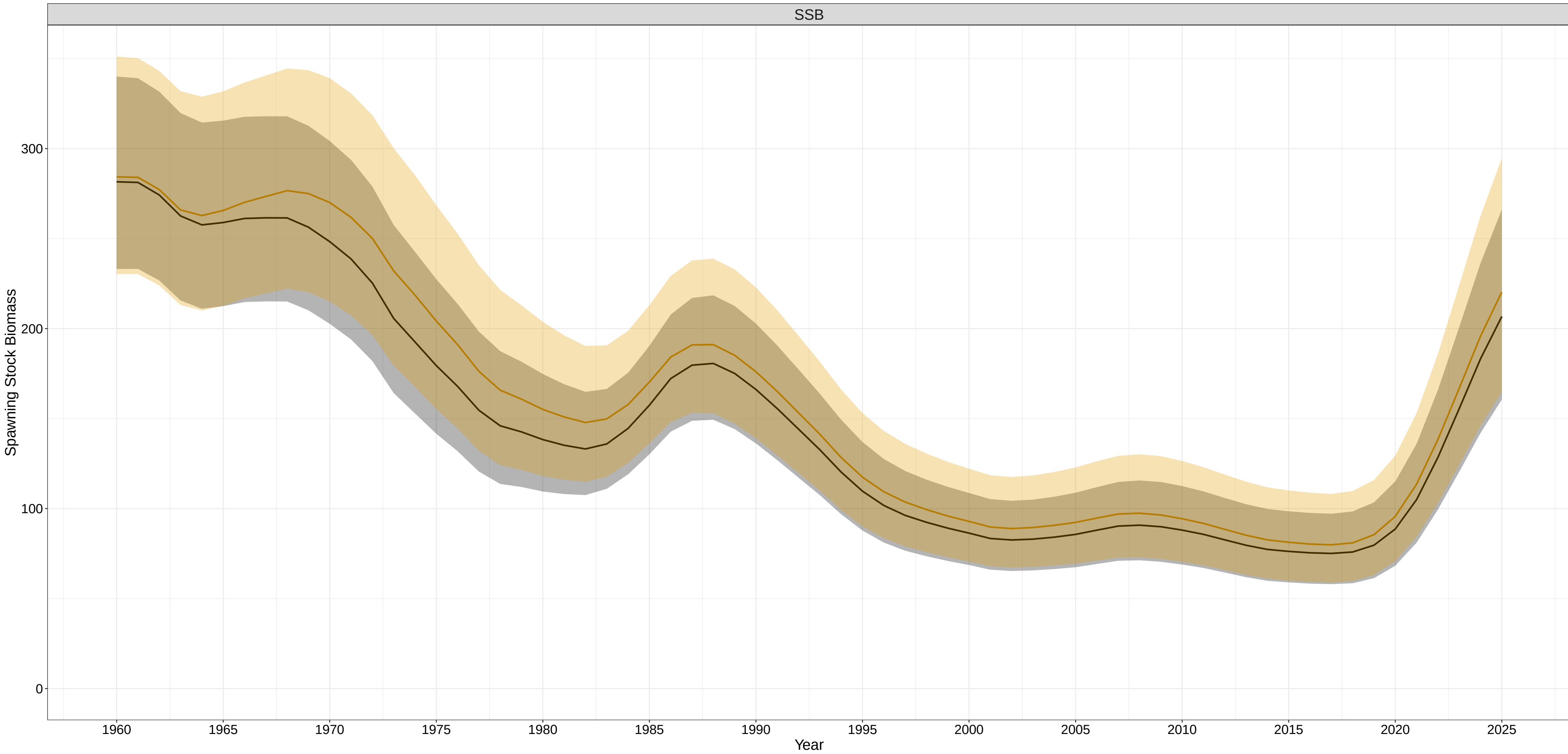
# Recruitment





SSB

Model 25.12\_1Reg\_Cont 25\_12\_Drop\_TS\_Upd\_M\_2025



# Biomass

